**P1.c**

```
#include<stdio.h>
 union A
 {
        struct st1
         {
                char c[2];   //2
                 char ch[2];  //2
        }s1;

        struct st2
        {
                 short int x;  //2
                 short int y;  //2
        }s2;

 }v={12,1,15,1};

 main()
 {
        printf("%d\n",v.s2.x);
         printf("%d\n",v.s2.y);
 }
```

**P2.c**

```
#include<stdio.h>

 union A

 {
         struct st1
        {
                char c[2];   //2
                char ch[2];  //2
         }s1;

        struct st2
        {
                short int x;  //2
                short int y;  //2
        }s2;

 }v={12,1,15,1};

 main()
 {
        printf("%d\n",v.s2.x);
        printf("%d\n",v.s2.y);
        v.s2.x=1026;
        v.s2.y=513;

         printf("%d\n",v.s1.c[0]);
        printf("%d\n",v.s1.c[1]);
         printf("%d\n",v.s1.ch[0]);
         printf("%d\n",v.s1.ch[1]);     }
```

**P3.c**

```
#include<stdio.h>
 union A
 {
         int x;
        char y;
        long long int z;
 };
 main()
 {
        union A v;
        printf("%d\n",sizeof(v));
}
```

**P4.c**

```
 #include<stdio.h>
  union A
  {
  int x;
  int y;
  }v={10};

 main()
 {
 printf("%d    %d\n",v.x,v.y);
 v.y=20;
 printf("%d    %d\n",v.x,v.y);
 }
```

**P5.c**

```
#include<stdio.h>
 union A
 {
         int x;
         int y;
         int z;
}v={10,20,30};

 main()
{
        printf("x add:%u\n",&v.x);
         printf("y add:%u\n",&v.y);
         printf("z add:%u\n",&v.z);

        printf("%d    %d   %d\n",v.x,v.y,v.z);
 }
```

**P6.c**

```
#include<stdio.h>
 union A
 {
         int x;
        char y;
 };

 main()
 {
        union A v={515};
         printf("v.x=%d  v.y=%d\n",v.x,v.y);

         v.y='a';

         printf("v.x=%d  v.y=%d\n",v.x,v.y);
 }
```

**P7.c**

```
 //Little Endian or Big Endian
 #include<stdio.h>
 union A
 {
         int x;
        char y;
 };
 main()
 {
        union A v={1};

        if(v.y==1)
                printf("Little Endian..\n");
        else
                printf("Big Endian..\n");
}
```

**P8.c**

```
 //wap to print float binary
 #include<stdio.h>
 union A
 {
         int x;
        float y;
 };

 main()
{
         int i;
         union A v;
         v.y=7.5;
         for(i=31;i>=0;i--)
                  printf("%d",v.x>>i & 1);
 }
```